Manual Hands-on HLA Sequence Analysis

Participant-facing instructions. Keep a scratchpad (Notepad/TextEdit) open throughout.

Step 1 - From genomic DNA to exons (CDS) to protein

Links (open in browser)

```
ANHIG/IMGTHLA (GitHub): https://github.com/ANHIG/IMGTHLA
SFAT (Sequence Feature Annotation Tool):
https://www.ebi.ac.uk/ipd/imgt/hla/annotation/
ExPASy Translate (optional): https://web.expasy.org/translate/
```

What you will produce

- your_allele_genomic.txt
- your_allele_CDS.txt
- your_allele_protein.txt

1.1 Download the genomic DNA sequence (FASTA)

Goal: start with the full genomic sequence (exons + introns) copied from a FASTA file.

What you do

1. Open the ANHIG/IMGTHLA repository in your browser (GitHub).
2. Go to the `fasta/` folder and open the genomic FASTA file for your locus.
   Genomic files typically follow this naming: `X_gen.fasta` where X = locus/gene (e.g., `A_gen.fasta`, `DRB1_gen.fasta`, `MICB_gen.fasta`).
3. To see the content of the file, press "`View raw`". Alternatively, press the Raw button, copy raw file button, or download button on the right.

4. Find your allele entry in the file: use `Ctrl+F` and search for the allele name (e.g., A*01:01:01:01).
5. Understand the FASTA layout: one header line starting with '>' (identifier + allele name + sometimes length/metadata), followed by the DNA sequence on the next lines (until the next entry or the end of the file).
   ```
   >HLA:HLA00001 A*01:01:01:01 3503 bp
   CAGGAGCAGAGGGGTCAGGGCGAAGTCCCAGGGCCCC
   ```
6. Copy the sequence letters only (A/C/G/T). **DO NOT** copy the header line that starts with '>'.
7. Open Notepad (or similar) as your scratchpad.

8. Save what you copied as your_allele_genomic.txt.

COPY / PASTE / SAVE box

COPY: DNA letters only (A/C/G/T) from the FASTA entry.

DO NOT COPY: the header line starting with '>' (gene/allele metadata).

PASTE: into Notepad first (scratchpad).

SAVE: as your_allele_genomic.txt

What you have now

- One long genomic DNA sequence (exons + introns) for one allele saved in a local txt file

1.2 Convert genomic DNA to coding DNA sequences with the Sequence Feature Annotation Tool

Goal: The Sequence Feature Annotation Tool (SFAT) identifies exon/intron features and outputs the coding sequence (CDS), which is the exonic sequence with introns removed.

Tool to use

- IPD-IMGT/HLA Sequence Feature Annotation Tool (SFAT).

What you do (click-by-click)

9. Open the Sequence Feature Annotation Tool page.
10. Paste your genomic DNA letters only (no FASTA header, no spaces).
11. Select the locus that matches your sequence (e.g., A, B, DRB1, MICB, etc.).
12. Select Level: `Genomic` (because your input includes both the exons and the introns).
13. Run the annotation by pressing the "`Annotate Sequence Now`" button.

COPY / PASTE / SAVE box

COPY: the full content of your_allele_genomic.txt

PASTE: into SFAT (letters only; no header).

SELECT: correct locus and Level = Genomic.

SAVE: copy the CDS output into your_allele_CDS.txt

WRITE DOWN: exon/intron boundaries and codon_start (if shown).

What to look for in the results

- Exon/intron boundaries (where each exon starts/ends in your input).
- A coding sequence (CDS): exonic sequence stitched together (introns removed).
- A codon_start / reading frame hint (important for correct translation).

1.3 Translate the CDS into a protein sequence

Goal: convert the CDS into amino acids (protein) so you can do downstream steps (alignment, surface accessibility, structure).

Option A (preferred): take the protein directly from SFAT output

14. In the SFAT results, find the Protein sequence in the "Protein" tab or at the bottom or in the GenBank/ENA/IMGT/HLA section.
15. Copy the one-letter amino-acid sequence into your_allele_protein.txt.

COPY / PASTE / SAVE box

COPY: the one-letter amino-acid sequence from the SFAT protein/translation output.

SAVE: as your_allele_protein.txt

Option B (manual/universal): translate the CDS with ExPASy Translate

16. Open ExPASy Translate (https://web.expasy.org/translate/).
17. Paste the CDS (from your_allele_CDS.txt) — NOT the genomic sequence.
18. Choose an output format (default is Compact: M, -, no spaces).
19. Press the "TRANSLATE!" button.
20. By default ExPASy shows 6 possible frames (3 forward, 3 reverse).
21. If SFAT provides codon_start, use it to select the correct forward frame:
    start=1 (in SFAT) -> frame 1 (in ExPASy); start=2 (in SFAT) -> frame 2 (in ExPASy);
    start=3 (SFAT)-> frame 3 (in ExPASy).
22. Copy the relevant protein sequence into your_allele_protein.txt.
23. Optional: ExPASy can download all frames as one text file.

COPY / PASTE / SAVE box

COPY: the full content of your_allele_CDS.txt

PASTE: into ExPASy Translate (not genomic DNA).

CHOOSE: the correct frame (use SFAT codon_start if available).

SAVE: as your_allele_protein.txt

Quick sanity checks (20 seconds)

- Did you accidentally paste the FASTA header (>...)? SFAT expects sequence letters only.
- Did you translate genomic DNA instead of CDS? That often creates many stop codons (*) because introns break the reading frame.
- Do you have a codon_start from SFAT? Use it; it prevents frame mistakes.

Step 2 - Analyse a synthetic DNA sequence of a 'novel allele'

Links (open in browser)

```
SFAT: https://www.ebi.ac.uk/ipd/imgt/hla/annotation/
Basic Local Alignment Search Tool (BLAST):
https://blast.ncbi.nlm.nih.gov/Blast.cgi
ExPASy Translate (optional): https://web.expasy.org/translate/
NCBI/NIH BLAST (optional): https://blast.ncbi.nlm.nih.gov/Blast.cgi
ANHIG/IMGTHLA GitHub (protein FASTA files): https://github.com/ANHIG/IMGTHLA
UniProt Align tool: https://www.uniprot.org/align
```

What you will produce

- synthetic_input_sequence.txt
- synthetic_CDS.txt
- synthetic_protein.txt
- closest_match_notes.txt
- reference_protein.fasta (closest matched allele, in FASTA format)
- synthetic_protein.fasta (your SFAT-derived protein, in FASTA format)
- uniprot_alignment.txt (copied alignment result + your 1-2 sentence interpretation)

BONUS:

- Did you know you can also query the IPD-IMGT/HLA Database via the website and get the sequences in different formats. Visit https://www.ebi.ac.uk/ipd/imgt/hla/alleles/ and search for an allele. Then see if you can find how to view the sequence in different formats.
- If you are familiar with them, there is also an API to help with this, take a look at https://www.ebi.ac.uk/ipd/imgt/hla/alleles/help/ for more info.

2.1 Clean the synthetic DNA sequence and save it

Goal: you start with a provided synthetic DNA sequence. Before analysis, make sure it contains only DNA letters (A/C/G/T) and nothing else.

What you do

24. Copy a synthetic DNA sequence from the workshop materials (GitHub).
25. Open Notepad (or similar) and paste the sequence.
26. Clean the sequence: remove spaces, line numbers, headers, and any other characters.
27. Keep only A/C/G/T.

28. Save the cleaned sequence as synthetic_input_sequence.txt.

COPY / PASTE / SAVE box

COPY: the full synthetic DNA sequence from the workshop materials.

PASTE: into Notepad (scratchpad).

CLEAN: remove spaces, line numbers, headers, and any non-ACGT characters.

SAVE: as synthetic_input_sequence.txt

What you have now

- One clean DNA sequence (A/C/G/T only) that you can safely paste into web tools.

2.2 Convert synthetic DNA to exons/CDS (and often protein) with SFAT

Goal: analyse the synthetic DNA with SFAT so you can identify exon/intron features (if present), generate the CDS (exons stitched together), and obtain a correct protein sequence.

Tool to use

- IPD-IMGT/HLA Sequence Feature Annotation Tool (SFAT).

What you do (click-by-click)

29. Open the Sequence Feature Annotation Tool page.
30. Paste your genomic DNA letters only (no FASTA header, no spaces).
31. Select the locus that matches your sequence (e.g., A, B, DRB1, MICB, etc.).
32. Select Level: `Genomic` (because your input includes both the exons and the introns).
33. Run the annotation by pressing the "`Annotate Sequence Now`" button.

COPY / PASTE / SAVE box

COPY: the full content of synthetic_input_sequence.txt

PASTE: into SFAT (letters only; no header).

SELECT: locus + correct Level (Genomic or CDS).

SAVE: copy CDS output into synthetic_CDS.txt

SAVE: if SFAT provides translation output, copy protein into synthetic_protein.txt

WRITE DOWN: locus selected + level selected + codon_start (if shown) in closest_match_notes.txt

What to look for in the results

- Exon/intron boundaries (where each exon starts/ends in your input). Note: splice changes can introduce premature stops and cause null alleles.
- A coding sequence: exonic sequence stitched together (introns removed).
- A codon_start / reading-frame hint (important for correct translation).
- The translated protein sequence.

2.3 Translate the CDS into protein (if not provided by SFAT)

Goal: ensure you end Step 2 with a correct one-letter amino-acid sequence so you can run protein-level comparisons.

Option A (preferred): take the protein directly from SFAT output

34. In the SFAT results, find the Protein section.
35. Copy the one-letter amino-acid sequence into synthetic_protein.txt.

COPY / PASTE / SAVE box

COPY: the one-letter amino-acid sequence from the SFAT translation output.

SAVE: as synthetic_protein.txt

Option B (manual/universal): translate the CDS with ExPASy Translate

36. Open ExPASy Translate.
37. Paste the CDS (from synthetic_CDS.txt) — not the genomic DNA.
38. Choose an output format (compact/continuous sequence).
39. ExPASy shows 6 possible frames (3 forward, 3 reverse).
40. If SFAT provides codon_start, use it to pick the correct forward frame:
    start=1 (in SFAT) -> frame 1 (in ExPASy);
    start=2 (in SFAT) -> frame 2 (in ExPASy);
    start=3 (SFAT)-> frame 3 (in ExPASy).
41. Copy the protein sequence into synthetic_protein.txt.
42. Optional: download all frames as a text file.

COPY / PASTE / SAVE box

COPY: the full content of synthetic_CDS.txt

PASTE: into ExPASy Translate.

CHOOSE: the correct frame (use SFAT codon_start).

SAVE: as synthetic_protein.txt

2.4 Find the closest known allele (NIH BLAST)

Goal: determine which known allele your synthetic sequence most closely resembles, and document why it looks novel relative to that closest match (identity, coverage, mismatches).

Preferred tools to use

- Basic Local Alignment Search Tool: https://blast.ncbi.nlm.nih.gov/Blast.cgi
- Alternative for HLA: There are lots of different BLAST services out there to use, but these may or may not have the same library of sequences to compare against or give the same results. Why not repeat the step above using the service at the EBI, https://www.ebi.ac.uk/jdispatcher/sss/ncbiblast? How do the results differ, why might this be?
- For DNA sequences: Nucleotide -> Nucleotide BLAST
- For protein sequences: Protein BLAST

What you do (Nucleotide similarity)

43. Go to the BLAST website: https://blast.ncbi.nlm.nih.gov/Blast.cgi (https://www.ebi.ac.uk/jdispatcher/sss/ncbiblast/nucleotide)
44. Your sequence contains DNA sequences, thus select the option: Nucleotide BLAST nucleotide -> nucleotide
45. Enter your Query Sequence in the text box.
46. Press the blue BLAST button on the lower left of the page and wait.
47. The output contains a list of sequences that are (partially) identical to your sequence, sorted by percent identity.
48. The closest matching HLA variant is on the top of that list.
49. You can inspect the homology in an alignment by clicking on the description.
50. It may make sense to select the one that has been described as “Complete CDS”.
51. Identical nucleotides are connected by a “|” between your query sequence and the obtained homologous subject. For mismatches, this symbol is missing.

COPY / PASTE / SAVE box

COPY: sequence from the appropriate file (genomic).

PASTE: into the matching NIH BLAST page.

RUN: keep defaults unless instructed otherwise.

SAVE: in closest_match_notes.txt record the top 3 hits for each run: allele name, % identity, alignment length/coverage, mismatches/gaps.

WRITE: 1–2 sentences: closest known allele + evidence + novelty evidence (differences at ...).

2.5 Align your synthetic protein with the closest matched allele (UniProt Align)

Goal: retrieve the protein sequence of the closest matched allele (the reference) from the ANHIG/IMGTHLA GitHub repository, and save it as a single FASTA entry.

What to do

52. Open the ANHIG/IMGTHLA repository in your browser.
53. Open the folder: `fasta/`
54. Open the protein file for your locus: X_prot.fasta (where X is the gene/locus, e.g., A_prot.fasta, DRB1_prot.fasta, MICB_prot.fasta). View the raw data file.
55. Find your closest matched allele entry in the file (use `Ctrl+F`).
56. Copy the entire FASTA entry: the header line starting with '>' **AND** the amino-acid sequence lines below it.

COPY / PASTE / SAVE box

COPY: Copy the full FASTA entry for the closest matched allele from X_prot.fasta (including the '>' header).

PASTE: Paste into Notepad/TextEdit as your scratchpad.

CHECK: You should see exactly one header line starting with '>' followed by only amino-acid letters (no numbers, no spaces).

SAVE: Save as reference_protein.fasta

What you have now

• A reference protein sequence for the closest matched allele, stored as FASTA (header + sequence) file.

2.6 Convert your SFAT protein output into FASTA format

Goal: your SFAT protein file often contains only the amino-acid letters. UniProt Align expects FASTA formatting, so you will add a header line and save it as a FASTA entry.

What to do

57. Open synthetic_protein.txt (your protein sequence from SFAT).
58. Create a new text file in Notepad/TextEdit.

59. Add a FASTA header line on the first line, starting with '>' (example: >synthetic_novel_allele).
60. Paste your protein sequence on the next lines (letters only; line wrapping is optional).
61. Save as synthetic_protein.fasta

COPY / PASTE / SAVE box

COPY: Copy the amino-acid letters from synthetic_protein.txt

PASTE: Paste into a new file under a header line like: >synthetic_novel_allele

CHECK: The file must contain two things only: one '>' header line and amino-acid letters (no '*', no spaces).

SAVE: Save as synthetic_protein.fasta

What you have now

• A FASTA-formatted protein entry for your synthetic allele.

2.7 Align the two proteins with the UniProt Align web tool

Goal: generate a protein alignment between your synthetic allele protein and the closest matched allele protein, so you can see the exact amino-acid differences in context.

What to do (click-by-click)

62. Open the UniProt Align tool in your browser.
63. In the sequence input box, paste BOTH FASTA entries, one after the other:
64. First: your synthetic protein FASTA (synthetic_protein.fasta)
65. Second: the reference protein FASTA (reference_protein.fasta)
66. Keep the default settings unless the instructor tells you otherwise.
67. Click Run Align.
68. In the results, locate the positions where the sequences differ (mismatches, gaps).
69. Copy the alignment output (or download if available) and save it for your notes.

COPY / PASTE / SAVE box

COPY: Copy the full FASTA text from synthetic_protein.fasta and reference_protein.fasta

PASTE: Paste both into the UniProt Align input box (two FASTA entries in one box).

RUN: Click Run Align.

SAVE: Copy the alignment result into uniprot_alignment.txt. Add 1–2 sentences: what is the closest match, and which residues differ.

What to look for in the results

Are there gaps? (Often indicates an insertion/deletion relative to the reference).

Are the differences clustered (e.g., ARD region) or scattered?

Do you see any '*' stop symbols? If yes, your input may contain a stop codon or a translation issue.

Quick sanity checks (worth 20 seconds)

- SFAT fails or gives nonsense features: you probably pasted spaces/headers or picked the wrong locus/level.
- Translation shows many stop codons (*): you translated genomic DNA instead of CDS, or selected the wrong frame.
- BLAST hit looks short: check coverage/alignment length; you may have pasted only part of the sequence.
- Did you paste FASTA headers for BOTH sequences? Each sequence must start with a line beginning with '>'.
- Did you accidentally paste DNA instead of protein? Protein should not contain A/C/G/T-only patterns and should not contain long stretches of only four letters.
- Are you aligning proteins from the same locus? Aligning A vs B will give nonsense.
- If the alignment looks shifted: re-check for missing residues at the start (signal peptide / leader) and confirm you copied the full entry from X_prot.fasta.

Output you will reuse in Step 3

Use the alignment to update your amino-acid variant list (position + AA change). This becomes the input for surface-exposure and antibody-relevance checks in Step 3.

Save the following results:

- synthetic_input_sequence.txt (DNA)
- synthetic_CDS.txt (CDS)
- synthetic_protein.txt (protein)
- Closest known allele from this step (from IPD-IMGT/HLA BLAST)

Step 3 - Is immune recognition plausible? (surface exposure + ARD + structural view)

Links (open in browser)

```
SFAT (Sequence Feature Annotation Tool):
https://www.ebi.ac.uk/ipd/imgt/hla/annotation/
IPD-IMGT/HLA Alignment tool: https://www.ebi.ac.uk/ipd/imgt/hla/alignment/
IPD-IMGT/HLA Allele Query tool: https://www.ebi.ac.uk/ipd/imgt/hla/alleles/
NetSurfP-2.0 (surface accessibility):
https://services.healthtech.dtu.dk/services/NetSurfP-2.0/
NCBI ORFfinder: https://www.ncbi.nlm.nih.gov/orffinder/
HLA Eplet Registry (optional): https://www.epregistry.com.br/
SpliceAI Lookup (optional): https://spliceailookup.broadinstitute.org/
RCSB PDB: https://www.rcsb.org/
PyMOL FindSurfaceResidues helper (optional):
https://pymolwiki.org/FindSurfaceResidues
```

What you will produce

- step3_variant_list.txt
- step3_truncation_check.txt
- step3_ard_check.txt
- step3_surface_check.txt
- Optional: step3_pymol_notes.txt

Inputs you need from previous steps

- synthetic_input_sequence.txt (DNA)
- synthetic_CDS.txt (CDS)
- synthetic_protein.txt (protein)
- Closest known allele from Step 2 (from IPD-IMGT/HLA BLAST)

3.0 Create your variant list (you will reuse it everywhere)

Goal: list each amino-acid difference between your synthetic protein and the closest known allele protein in a clean, consistent format.

What you do

70. Open synthetic_protein.txt.
71. Compare it to the closest known allele protein (from BLAST/alignment).
72. Write down each amino-acid difference in a clean, consistent way.

COPY / PASTE / SAVE box

COPY: your synthetic protein sequence (and the closest allele protein sequence or alignment snippet).

WRITE (format): one change per line, e.g., 80 I->T; 116 D->Y; 156 insertion: +G; 233 deletion: ΔK

SAVE: as step3_variant_list.txt

Quick rule

- If you have only DNA differences: do not guess. First translate (SFAT/ORFfinder/ExPASy), then list AA changes.

3.1 Is this variation leading to a premature end of the protein?

3.1.1 Stop codon introduced by the variant

Goal: check if the variant introduces an early stop (truncation).

What you do

73. Open synthetic_protein.txt.
74. Look for an early stop sign (some tools show '*' as stop) or an unexpectedly short protein.
75. Cross-check with NCBI ORFfinder (fast sanity check).

COPY / PASTE / SAVE box

COPY: synthetic_CDS.txt (preferred) or synthetic_input_sequence.txt (if CDS not available).

PASTE: into ORFfinder and run.

LOOK FOR: the longest ORF; whether it stops much earlier than expected; whether multiple ORFs appear because of early stops.

SAVE: write a 3–5 line conclusion in step3_truncation_check.txt: 'Stop codon present: yes/no' + evidence (protein has */ORF ends at ...).

3.1.2 Frameshift (insertion/deletion not divisible by 3)

Goal: check whether an indel causes a frameshift and therefore likely truncation.

What you do

76. From your coding alignment, identify insertions/deletions in the coding sequence.
77. Check the length: 3, 6, 9... bases -> in-frame; 1, 2, 4, 5... bases -> frameshift.

78. Confirm with ORFfinder: a frameshift often creates a short, abnormal ORF.

COPY / PASTE / SAVE box

COPY: the exact indel from your alignment (length and position).

WRITE: in step3_truncation_check.txt add: 'Frameshift: yes/no', 'Indel length: ... bp', 'Expected consequence: early stop likely / not likely'.

3.1.3 Alternative splicing / splice site disruption (genomic sequences)

This only applies if your synthetic sequence includes introns (genomic input).

What you do (simple, manual but effective)

79. Run SFAT on the genomic sequence and record exon boundaries.
80. Check whether your variant sits at an exon boundary:
81. - Donor site (start of intron): usually GT right after the exon
82. - Acceptor site (end of intron): usually AG right before the next exon
83. If your variant changes those canonical bases, splicing disruption is plausible.

Optional (stronger): SpliceAI Lookup

If you can express your change as a variant on a reference genome coordinate, use SpliceAI Lookup and record the delta score.

COPY / PASTE / SAVE box

WRITE: in step3_truncation_check.txt add: 'Splice-site hit: yes/no', 'Which boundary: exon X donor/acceptor', 'Reason: canonical GT/AG changed / nearby splice motif changed'.

OPTIONAL: add 'SpliceAI delta score' if used.

3.2 Does this change the Antigen Recognition Domain (ARD; and therefore G/P grouping logic)?

ARD definition used here

- Class I: peptide-binding domain is encoded by exons 2 and 3.
- Class II: peptide-binding domain is encoded by exon 2.

3.2.1 Does this lead to another G-group characterization?

Goal: determine whether the ARD nucleotide sequence differs from the closest allele (relative G-group statement).

What you do

84. Using SFAT output, isolate the ARD nucleotide sequence:
85. - Class I: exon 2 + exon 3
86. - Class II: exon 2
87. Compare that ARD nucleotide sequence to the closest allele's ARD nucleotide sequence (IPD alignment tool can show exon-level alignments).
88. If any nucleotide differs in the ARD region, it is not the same G-group as the closest allele (relative statement).

COPY / PASTE / SAVE box

COPY: ARD exon nucleotide sequence(s) from SFAT + closest allele ARD exon sequence(s).

COMPARE: identical yes/no.

SAVE: in step3_ard_check.txt write: 'ARD nucleotide change: yes/no' and 'Same G-group as closest allele? yes/no (relative)'.

3.2.2 Does this lead to another P-group characterization?

Goal: determine whether the ARD protein sequence differs from the closest allele (relative P-group statement).

What you do

89. Translate the ARD region (or use your protein alignment) and check whether any AA in the ARD differs.
90. Any ARD amino-acid difference vs closest allele implies a different P-group (relative statement).

COPY / PASTE / SAVE box

COPY: your ARD AA changes from step3_variant_list.txt.

DECIDE: are any within ARD? (Class I: exons 2+3; Class II: exon 2).

SAVE: in step3_ard_check.txt write: 'ARD protein change: yes/no' and 'Same P-group as closest allele? yes/no (relative)'.

3.2.3 Does this alter the peptide repertoire (binding groove)?

You do not prove this here — you flag plausibility.

What you do (workshop rule)

91. If the change is in ARD, it may affect peptide repertoire.
92. Strengthen your conclusion using structure (Step 3.3 + optional PyMOL): if the mutated residue points into the binding groove or sits close to the peptide, plausibility increases.

COPY / PASTE / SAVE box

WRITE: in step3_ard_check.txt: 'Peptide repertoire impact plausible: low / medium / high'.

WRITE: reason: 'ARD change yes/no; residue points to groove yes/no'.

3.3 Is this variation surface-accessible (antibody/NK/T cell relevance)?

You will depict exposure with two options:

- Prediction from sequence (NetSurfP-2.0).
- Structure view using a PDB reference in PyMOL (optional).

3.3.1 Run the surface accessibility analysis (NetSurfP-2.0)

Goal: obtain per-residue exposure (RSA) predictions for your synthetic protein.

What you do

93. Open NetSurfP-2.0.
94. Paste synthetic_protein.txt.
95. Run the job.
96. Download/export the per-residue output.

COPY / PASTE / SAVE box

COPY: protein sequence from synthetic_protein.txt

PASTE: into NetSurfP and run.

SAVE: export/download results and copy the key lines (your variant positions) into step3_surface_check.txt.

3.3.2 Identify exposure characteristics of the new amino acids

Goal: map each AA change to exposure (exposed/buried) using NetSurfP output.

What you do

97. For each line in step3_variant_list.txt, look up the same position in the NetSurfP output.
98. Record whether it looks exposed or buried (use RSA as your guide).

COPY / PASTE / SAVE box

SAVE: in step3_surface_check.txt create one block per variant:

- Position + AA change
- RSA (from NetSurfP)
- **Call:** exposed / partly exposed / buried
- 1-line interpretation: 'Accessible change yes/no'

3.3.3 Could this lead to antibody recognition by Bw4/Bw6?

Goal: check whether the variant affects the class I 77–83 motif (Bw4/Bw6 region).

What you do

99. Check if any variant position falls within residues 77–83 (class I only).
100. If yes: write down the 77–83 motif for your synthetic allele and compare it to the closest allele's motif.
101. If the motif changes, Bw4/Bw6 reactivity can change.

COPY / PASTE / SAVE box

SAVE: in step3_surface_check.txt write: '77–83 affected: yes/no', 'Motif changed: yes/no', 'Bw4/Bw6 relevance: plausible / unlikely'.

3.3.4 Could this lead to antibody recognition in general?

Goal: do a useful screen without overclaiming specific antibody specificities.

What you do (workshop-level decision rule)

102. If a residue is surface-accessible (NetSurfP) and different from the closest allele, it can be part of an antibody epitope.
103. Optional stronger check: see if the position overlaps a known eplet in the HLA Eplet Registry.

COPY / PASTE / SAVE box

COPY: variant positions that are 'exposed' from NetSurfP mapping.

OPTIONAL CHECK: look up those positions in the HLA Eplet Registry.

SAVE: in step3_surface_check.txt add: 'Antibody-relevant variants (exposed): [list]', 'Eplet overlap: yes/no/unknown', 'Overall antibody recognition plausibility: low/medium/high'.

3.3.5 Could this lead to altered NK recognition?

Goal: flag plausibility of NK receptor effects.

What you do

104. If your variant affects residues 77–83 (class I), NK impact via the Bw4 axis becomes more plausible.
105. If not, you can still note 'unknown/possible', but do not overclaim.

COPY / PASTE / SAVE box

SAVE: in step3_surface_check.txt write: 'KIR/Bw4 axis affected: yes/no' and 'NK recognition change plausible: low/medium/high' + 1-line reason.

3.3.6 Could this lead to altered direct T cell recognition?

Goal: flag plausibility based on ARD involvement and exposure (TCR-facing surface).

What you do

106. If your change is in ARD and exposed, direct T cell recognition impact is plausible.
107. Strengthen with PyMOL: if residue is on the TCR-facing side, plausibility increases.

COPY / PASTE / SAVE box

SAVE: in step3_surface_check.txt write: 'Direct T cell recognition change plausible: low/medium/high' and 'Reason: ARD change yes/no; exposed yes/no'.

Step 3 final deliverable (max 5 bullets)

- Premature end? yes/no + reason (stop/frameshift/splice).

- ARD affected? yes/no.
- G/P group change relative to closest allele? yes/no (relative statement).
- Accessible polymorphism? yes/no (NetSurfP +/- PyMOL).
- Antibody/NK/T cell plausibility: low/medium/high + 1-line reason (mention Bw4/Bw6 if relevant).

Step 4 - Map your changed amino acids onto an HLA structure (web-based, using RCSB Mol*)

Example: HLA-A*02:01

What you will produce

- step4_pdb_model.txt (which PDB you used + why)
- step4_structure_mapping.txt (per variant: exposed/buried + notes)
- Optional: a folder step4_screenshots/ (PNG screenshots exported from Mol*)

Inputs you need from earlier steps

- step3_variant_list.txt (your amino-acid changes, positions)
- The closest known allele from Step 2
- For this example, you will use an HLA-A*02:01 PDB entry

Links (open in browser)

```
RCSB PDB (search and 3D viewer):
https://www.rcsb.org/

Mol* getting started (RCSB documentation):
https://www.rcsb.org/docs/3d-viewers/mol*/getting-started
```

4.1 Find a proper HLA-A*02:01 PDB model (RCSB)

Goal

Pick a structure that is usable for surface mapping. You want at least:

- HLA heavy chain (HLA-A*02:01)
- beta2-microglobulin
- a bound peptide (preferred, because it orients the groove)

What to do

1. Open RCSB PDB (browser).
2. Search for: HLA-A*02:01
3. Open a candidate structure page and verify it is HLA-A*02:01.
4. Check that the structure contains beta2-microglobulin and a bound peptide.

5. Prefer X-ray or cryo-EM structures (defined coordinates).
6. Pick one PDB entry as your reference model.

Examples of PDB entries that work for HLA-A*02:01 (examples only)

- 7M8S (HLA-A*02:01 with peptide; useful for groove orientation)
- 5F9J (HLA-A*02:01 with peptide)
- If you need an antibody/Fab present (optional), choose a pMHC structure that includes a Fab chain

COPY / PASTE / SAVE box

COPY: PDB ID, title (1 line), and which components are present (heavy chain / beta2m / peptide / extras).

SAVE: Create step4_pdb_model.txt with: Chosen PDB, Reason, Notes (extras present yes/no; method/resolution optional).

4.2 Open the structure in Mol* and clean the view

Goal

Get a clear picture: HLA heavy chain surface plus your residues highlighted.

What to do

1. On the RCSB PDB page, open the 3D View (Mol*).
2. Locate the Sequence panel and the Components/Controls panel.
3. Keep visible: HLA heavy chain, beta2-microglobulin, and peptide.
4. Hide: any extra chains (Fab/TCR/other) if present.

Important

Mol* works by selecting parts of the structure and then applying actions to that selection (color, transparency, representations, or making a separate component).

COPY / PASTE / SAVE box

WRITE: In step4_structure_mapping.txt add: PDB used, Viewer (Mol*), and which objects you kept visible.

4.3 Prepare the residue positions you need to map

Goal

Have a clean list of residue numbers you must highlight.

What to do

1. Open step3_variant_list.txt.
2. Extract only the residue numbers (positions).
3. Write them as a comma-separated list.

COPY / PASTE / SAVE box

COPY: Residue positions from step3_variant_list.txt.

WRITE: In step4_structure_mapping.txt add a line: Variant positions to map: 80, 116, 156, ...

RULE: If PDB residue numbering does not match your sequence numbering, document the mismatch (see sanity checks).

4.4 Select the changed residues in Mol*

Goal

Create a selection that contains all your variant residues.

4.4.1 Turn on Selection Mode and set Picking Level to Residue

1. Click the cursor icon to enable Selection Mode.
2. Set Picking Level to Residue (so you select whole amino acids, not single atoms).
3. Click a residue once to confirm it becomes highlighted.

CHECK: When you click a residue, it highlights in the 3D view and in the sequence panel.

4.4.2 Select residues from the Sequence panel

1. In Selection Mode, click the residue in the sequence panel at the correct position.
2. To select a range, hold Shift while clicking along the sequence.
3. To select multiple separate residues, use Add/Union selection and click each residue one by one until all are included.

GOAL: You end with one selection that contains all variant residues.

TIP: Use Add/Union selection to accumulate non-adjacent residues.

4.4.3 Select residues by clicking in the 3D canvas

1. Rotate the structure until the region is visible.
2. Click the residue on the 3D model.
3. Use Add/Union selection for multiple residues (same as above).

4.5 Color your selected residues so they stay visible

Goal

Make the mutated residues stand out.

What to do

1. With your residues selected, click Apply Theme to Selection.
2. Choose a strong color and apply it.
3. Confirm that the selection remains visible after you click elsewhere.

WRITE: In step4_structure_mapping.txt add: Color applied to variants: yes/no; How: Apply Theme to Selection (Mol*).

4.6 Show the HLA heavy chain as a surface (exposed vs buried)

Goal

Decide if each polymorphism is accessible to antibody binding.

What to do

1. Ensure the HLA heavy chain is visible.
2. Change the representation to include a molecular surface for the heavy chain component.
3. Optional: set the surface slightly transparent so you can still see colored sticks underneath.

DECIDE (for each residue): Exposed (visible on surface), Buried (inside), or Unclear (ambiguous).

SAVE: In step4_structure_mapping.txt create one block per variant: Position, Change, Surface call, Notes (1 line).

4.7 Optional: make a separate component for the variant residues (clean pictures)

Goal

Surface plus colored sticks is the clearest way to present results.

What to do

1. Select the variant residues.
2. Use Create Component of Selection with Representation.
3. Choose a representation like sticks.
4. Use the eye icon to show/hide the component.

4.8 Export a screenshot

What to do

1. Set the view: surface visible, residues colored, peptide visible if present.
2. Use the camera/screenshot option and download the image.

SAVE: Recommended filenames: step4_HLA-A0201_surface_variants.png; optional: step4_HLA-A0201_zoom_variant80.png

WRITE: In step4_structure_mapping.txt add: Screenshots saved yes/no; file names.

Quick sanity checks (do these before you conclude)

- Residue numbers do not exist in the PDB sequence panel: you may have a numbering offset or a trimmed construct. Pick a different PDB entry or document the offset in step4_pdb_model.txt.
- Extra chains block the view: hide them using the Components panel (eye icon), or keep only the pMHC components.
- Selected residues disappear after clicking elsewhere: apply the theme immediately or create a separate component from the selection.